Refinements to the SAE–Thompson Sampling Framework for Active Email Outreach

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1. Introduction

Žid et al. (2025) frame cold-start recipient selection for a new email campaign as an iterative multi-armed bandit: each arm is a recipient, and Thompson Sampling (TS) parameters are derived from a shallow autoencoder (SAE) trained on historical recipient–template opens, avoiding retraining during the active-learning phase. Several components of that model — a fixed exploration/exploitation coefficient, an unweighted historic open rate, a linear aggregation for the current-state confidence term, training on binary labels only, and a shallow architecture — were fixed by the original authors for simplicity. We revisit each of these choices and propose concrete, mathematically grounded refinements: dynamic scheduling of the blend coefficient, a recency-weighted historic rate, a nonlinear forward-pass confidence term, a variance-based influence score, a factored score function, two ways of exposing the autoencoder to previously unused time-to-open (TTO) signal, and a deep, nonlinear autoencoder. We report grid-search results for each proposal against a reproduced baseline and outline directions for further work.

2. Recap of the Original Model

Setup. We observe $X \in \{0,1\}^{n\times m}$, where n is the number of templates, m the number of recipients, and $X_{i,j} = 1$ iff recipient j opened template i. For a new template $n+1$, sent over b batches within a deadline T, the model selects at each batch a small subset of recipients so as to maximize eventual openers reached early.

Shallow autoencoder. A SAE with encoder/decoder $E, D \in \mathbb{R}^{m\times d}$ is trained as

$$\min_{E,D} \ell\big(X,\ \sigma(XB_{E,D})\big), \qquad B_{E,D} = ED^\top - \operatorname{diag}\big([E \odot D]\mathbf{1}_m\big),$$

with ℓ the element-wise binary cross-entropy and the diagonal constraint ruling out trivial self-prediction. We write $f_{SAE}(x) = \sigma(x^\top B_{E,D})$ for the trained forward pass and $\Sigma = \sigma(B_{E,D})$; entry $\Sigma_{i,j}$ is the probability of recipient j opening a template given that only recipient i has opened it — a measure of i's influence on j.

Arm score. The bandit score for recipient j at time t is

$$s_j(t) = \alpha\, \phi_j p_j + (1-\alpha) f_j(t), \tag{1}$$

$$\phi_j = \frac{1}{n}\sum_{i=1}^{n} X_{i,j}, \qquad p_j = \frac{1}{m-1}\sum_{i\neq j} \Sigma_{j,i}, \qquad f_j(t) = \bar{x}(t)^\top \Sigma_{:j}, \tag{2}$$

where ϕ_j is recipient j's historic open rate, p_j is a general influence score, $f_j(t)$ is a confidence score specific to template $n+1$ given the observed (normalized) state $\bar{x}(t)$, and $\alpha \in [0,1]$ trades off the two. TS parameters are then $\alpha_j(t) = G s_j(t)$, $\beta_j(t) = G(1 - s_j(t))$ for a confidence modifier G.

Two orthogonal readings of Eq. (1) motivate our refinements: $\phi_j p_j$ vs. $f_j(t)$ can be read as *exploration vs. exploitation*, or equivalently as *historic vs. current* data.

3. Proposed Refinements

3.1 Dynamic α-Scheduling

A fixed α ignores that the value of exploration should shrink as the campaign progresses. We schedule $\alpha(t)$ against two natural progress variables: $\mu = N(t)/N$, the fraction of the sending budget already used, and $\tilde{o} = o(t)/m$, the fraction of recipients who have already opened, with $o(t) = x(t)^\top \mathbf{1}$.

Send-volume-based (function of μ). A **linear** schedule from l to r,

$$\alpha(\mu) = l(1-\mu) + r\mu, \tag{3}$$

and a **geometric** (log-linear) schedule,

$$\alpha(\mu) = l^{1-\mu}\, r^{\mu}. \tag{4}$$

Open-rate-based (function of $\tilde{o}$). Since $f_j(t)$ is itself a point estimate that becomes more reliable as more opens are observed, we also consider a **confidence** schedule, hyperbolic in $\tilde{o}$:

$$\alpha(\tilde{o}) = \frac{\kappa}{\tilde{o} + \kappa}, \tag{5}$$

with dampening factor κ (so $\alpha(\kappa) = \frac{1}{2}$). All three reduce to the constant baseline at their degenerate limits ($l = r$, or $\kappa \to \infty$). A schedule jointly conditioned on both μ and $\tilde{o}$ is conceptually natural but is left to future work (Section 5) to avoid an unconstrained hyperparameter count.

3.2 Recency-Weighted Historic Rate ϕ_j

The original ϕ_j weights every past template equally, ignoring drift in recipient behavior over time. Assuming templates $X_{1:}, \ldots, X_{n:}$ were sent at uniform intervals, we assign template i an exponential weight

$$\omega_i = 2^{-d_i/h}, \qquad d_i = \frac{n-i}{n}, \tag{6}$$

with half-life $h > 0$, and redefine

$$\phi_j = \frac{\omega^\top X_{:j}}{\sum_i \omega_i}. \tag{7}$$

Normalizing by $\sum_i \omega_i$ keeps $\phi_j \in [0,1]$. As $h \to \infty$ every $\omega_i \to 1$, recovering the original unweighted average as a special case.

3.3 Forward-Pass Definition of $f_j(t)$

$f_j(t) = \bar{x}(t)^\top \Sigma_{:j}$ is a linear combination of individual influences $\Sigma_{i,j}$ and therefore captures only the *average* effect of openers, never their joint interaction. We instead pass the full current state through the trained network:

$$f_j(t) = \sigma\big(x(t)^\top B_{E,D}\big) e_j = f_{SAE}(x(t))\, e_j. \tag{8}$$

Because σ is nonlinear, $\sigma(x(t)^\top B_{E,D}) \neq x(t)^\top \sigma(B_{E,D})$ in general (equality holds only for one-hot $x(t)$), so Eq. (8) is a genuinely different, jointly-conditioned probability estimate rather than a relabeling of Eq. (2).

3.4 Variance-Based Definition of p_j

p_j is meant to reward recipients whose opening is *informative* about others. The original definition, $p_j = \frac{1}{m-1}\sum_{i\neq j} \Sigma_{j,i}$, rewards recipients that raise the *average level* of other recipients' f-scores — which is not the same as reducing uncertainty about them. We instead reward maximal *spread* induced across other recipients' scores:

$$p_j = \frac{4}{m-1} \sum_{i\neq j} \big(\Sigma_{j,i} - \bar{\Sigma}_{j:}\big)^2, \qquad \bar{\Sigma}_{j:} = \frac{1}{m-1} \sum_{i\neq j} \Sigma_{j,i}. \tag{9}$$

This is the variance of $\Sigma_{j,:}$ (excluding j), scaled by 4 so that the maximum possible variance of a Bernoulli variable (0.25) maps to 1, keeping $p_j \in [0,1]$.

3.5 Factored Score $s_j(t) = \pi_j(t)\, u_j(t)$

The linear combination in Eq. (1) conflates the two orthogonal axes noted in Section 2 — it is not clear, for instance, why exploration should draw solely on historic data while exploitation draws solely on current data. We instead factor the score into a *probability-of-opening* term and a *utility-of-opening* term, each independently schedulable:

$$\pi_j(t) = \alpha(t)\phi_j + (1 - \alpha(t))f_j(t), \qquad u_j(t) = \beta(t)p_j + (1 - \beta(t)) \cdot 1, \tag{10}$$

$$s_j(t) = \pi_j(t)\, u_j(t) = \Big(\alpha(t)\phi_j + [1 - \alpha(t)]f_j(t)\Big)\Big(\beta(t)p_j + [1 - \beta(t)]\Big). \tag{11}$$

Here $\pi_j(t)$ blends historic and current opening-probability estimates, while $u_j(t)$ blends the direct value of an open (the constant 1) with its indirect information value p_j. This factorization reuses every primitive introduced above unchanged, so refinements from Sections 3.1–3.4 can be dropped into $\pi_j(t)$ and $u_j(t)$ directly — we use a confidence schedule (Eq. 5) for $\alpha(t)$ and a send-volume schedule (Eq. 3/4) for $\beta(t)$.

3.6 Incorporating Time-to-Open (TTO)

The SAE is trained purely on binary opens, discarding *how fast* a recipient opened — signal that matters given the short operational window $(T/b,\, T)$. Let $\Delta_{i,j} \in \mathbb{R}_{\geq 0} \cup \{+\infty\}$ be the observed time-to-open (with $\Delta_{i,j} = +\infty$ for non-openers, so $X_{i,j} = \mathbb{1}[\Delta_{i,j} < \infty]$), and define a decayed open label

$$Y_{i,j} = 2^{-\Delta_{i,j}/h_\delta}, \qquad 2^{-\infty} := 0, \tag{12}$$

with TTO half-life h_δ; as $h_\delta \to \infty$, $Y_{i,j} \to X_{i,j}$ for every opener.

Binary-to-TTO autoencoder. We keep the SAE's binary input but replace the reconstruction target:

$$\min_{E,D}\ \ell\big(Y,\ \sigma(XB_{E,D})\big). \tag{13}$$

Since the input space is unchanged, the forward-pass definitions of p_j and $f_j(t)$ (Eqs. 8–9) carry over mechanically; only their interpretation shifts — $\Sigma_{j,i}$ now blends *whether* i opens with *how fast*.

TTO cutoff. Training rows of X reflect the *eventual* open pattern, while active-learning queries $x(t)$ are necessarily censored at $t < T$. We instead censor the training input itself, fixing a cutoff δ_c and defining $C_{i,j} = \mathbb{1}[\Delta_{i,j} \leq \delta_c]$, then training

$$\min_{E,D} \ \ell\big(X,\ \sigma(CB_{E,D})\big). \tag{14}$$

C retains only opens fast enough to plausibly be observed within an active operational window and treats slower opens as non-opens during training — matching the censoring recipients are actually subject to during active learning. As $\delta_c \to \infty$, $C \to X$, recovering the original model.

3.7 Deep Autoencoder

$B_{E,D}$ in Eq. (2)/(2) is a single fixed $m \times m$ matrix, so the SAE can only capture linear recipient–recipient relationships. Since p_j and $f_j(t)$ (Section 3.3–3.4) already depend only on the ability to query f_{SAE}, not on any explicit property of $B_{E,D}$, we can swap in a deeper, nonlinear network without changing either definition. We use a two-layer encoder/decoder with a ReLU nonlinearity,

$$\text{Encoder: Linear}(m, 2d) \to \text{ReLU} \to \text{Linear}(2d, d), \qquad \text{Decoder: Linear}(d, 2d) \to \text{ReLU} \to \text{Linear}(2d, m), \tag{15}$$

giving $f_{DAE}(x) = (\sigma \circ \text{Decoder} \circ \text{Encoder})(x)$, optionally with dropout and bottleneck-layer normalization. A deep network has no single weight matrix whose diagonal can be constrained to zero; we substitute a denoising objective, randomly masking input entries and requiring reconstruction of the full unmasked X, forcing every prediction to depend on other recipients rather than a self-shortcut. Training becomes $\min_\theta \ell(X, f_{DAE}(X;\theta))$, and p_j, $f_j(t)$ retain their Section 3.3–3.4 forms with f_{SAE} replaced by f_{DAE}.

4. Experiments

4.1 Methodology

For each variant we grid-search its hyperparameters and select the configuration maximizing validation-set Recall-AUC, $\text{AUC} = \int_0^1 \text{Recall}(\tau)\, d\tau$, where $\text{Recall}(\tau)$ is recall at sending fraction τ. We report test-set Recall@5/15/25/35% and Recall-AUC, mean ± std over repeated simulations. Before testing modifications we reproduced the original model on our pipeline; at the operating points reported by Žid et al. (25/50/75%) our reproduction matches the published point estimates exactly (0.923, 0.975, 0.989), with larger variance attributable to fewer simulation runs.

4.2 Results

Variant	Selected hyperparameters	Recall@5%	Recall@15%	Recall@25%	Recall@35%	AUC
Baseline	—	0.385 ± 0.076	0.821 ± 0.028	0.923 ± 0.014	0.958 ± 0.011	0.894 ± 0.012
Linear α-schedule	$l = 0.1,\ r = 0.05$	0.383 ± 0.073	0.823 ± 0.023	0.927 ± 0.012	0.960 ± 0.010	0.894 ± 0.011
Geometric α-schedule	$l = 0.3,\ r = 0.05$	0.377 ± 0.050	0.823 ± 0.017	0.927 ± 0.012	0.959 ± 0.010	0.894 ± 0.009
Confidence α-schedule	$\kappa = 0.003$	0.341 ± 0.031	0.818 ± 0.022	0.926 ± 0.014	0.960 ± 0.010	0.891 ± 0.009
Recency-weighted ϕ_j	$h = 0.3$	0.447 ± 0.041	0.837 ± 0.018	0.928 ± 0.014	0.960 ± 0.011	0.903 ± 0.009
Forward-pass $f_j(t)$	—	0.375 ± 0.049	0.841 ± 0.029	0.937 ± 0.018	0.966 ± 0.010	0.901 ± 0.011
Factored $s_j(t)$	$\alpha(t)$: conf., $\kappa = 0.005$; $\beta(t)$: geom., $l = 0.1,\ r = 0.05$	0.425 ± 0.026	0.821 ± 0.017	0.922 ± 0.013	0.959 ± 0.010	0.900 ± 0.008
Variance-based p_j	—	0.479 ± 0.034	0.837 ± 0.017	0.926 ± 0.013	0.960 ± 0.010	0.905 ± 0.009
Binary-to-TTO autoencoder	$h_\delta = 11520$	0.331 ± 0.051	0.719 ± 0.040	0.896 ± 0.015	0.955 ± 0.008	0.877 ± 0.011
TTO cutoff	$\delta_c = 720$	0.390 ± 0.038	0.824 ± 0.016	0.924 ± 0.012	0.960 ± 0.010	0.895 ± 0.008
Deep autoencoder	$d = 16$	0.404 ± 0.086	0.829 ± 0.052	0.931 ± 0.019	0.963 ± 0.011	0.900 ± 0.017

4.3 Discussion

Differences from baseline are largest at low sending fractions and shrink as τ grows, which is expected: baseline recall already exceeds 0.92 by $\tau = 25\%$, leaving little headroom. The **variance-based** p_j (0.479 at 5%) and **recency-weighted** ϕ_j (0.447) give the largest low-fraction gains, followed by the **factored** $s_j(t)$ (0.425); all three also improve AUC (0.905, 0.903, 0.900 vs. 0.894 baseline). The **forward-pass** $f_j(t)$ shows no gain at 5% but a consistent uplift from 15% onward (0.841/0.937/0.966), suggesting nonlinear aggregation of openers only pays off once several recipients have opened. The three **α-scheduling** variants are essentially neutral at low τ, and the confidence schedule is *worse* than baseline (0.341 at 5%) — its effective range is likely too narrow given the dataset's low (9.1%) empirical open rate, and κ warrants re-tuning. The **Binary-to-TTO autoencoder** underperforms across every quantile, most severely at 15% (0.719 vs. 0.821): discounting slow opens in the reconstruction target appears to degrade collaborative-filtering signal quality more than the speed information adds, at least at the selected h_δ; this may also reflect the confidence modifier G (tuned for the original Σ) being miscalibrated for a TTO-blended Σ of different scale. **TTO cutoff** censoring gives a small, consistent uplift across all quantiles at negligible interpretive cost — training-time censoring that matches the active-learning observability window helps modestly without changing what Σ represents. The **deep autoencoder** improves Recall@5% (0.404) and the higher quantiles modestly, but with markedly higher variance throughout (e.g. ± 0.086 at 5%, ± 0.052 at 15% — 2–3× most other variants), indicating the added capacity is not yet well-regularized at this dataset size. All results test one modification at a time; combinations are untested.

5. Future Work

- **Combined α-scheduling.** A schedule jointly conditioned on send-progress μ and open-progress $\tilde{o}$ rather than either alone (Section 3.1); the main obstacle is the added hyperparameter count.
- **Plug-and-play composition under the factored score.** Systematically combine the primitive-level refinements — recency-weighted ϕ_j, forward-pass $f_j(t)$, variance-based p_j, TTO- or deep-autoencoder-derived Σ — inside $\pi_j(t)$ and $u_j(t)$ (Eq. 10) rather than evaluating each in isolation, and search jointly rather than ablating one factor at a time, since the gains in Section 4.3 are not guaranteed to be additive.
- **$z(t)$ interpolation for TTO-informed inference.** Rather than querying $f_j(t)$ on the binary $x(t)$, define a partially TTO-informed state $z_i(t) = 2^{-\delta_i/h_\delta}$ for recipients who have already opened template $n+1$ (and 0 otherwise), and evaluate $f_j(t) = f_{SAE}(z(t))e_j$. Since $x^\top B_{E,D}$ is linear and $z(t) \in [0,1]^m$, this is a valid extension of a model trained on binary rows; whether early-openers' TTO adds signal beyond the binary state is untested.
- **Hyperbolic decay for Y.** $Y_{i,j} = \kappa_\delta/(\Delta_{i,j} + \kappa_\delta)$ gives a heavier tail than the exponential label of Eq. (12) and merits a direct comparison.
- **TTO-to-TTO autoencoder.** Replacing the SAE input with Y as well, $\min_{E,D} \ell(Y, \sigma(YB_{E,D}))$, exposes TTO on both sides but changes the role of e_j — an instantaneous open becomes a boundary point rather than

a typical input — so p_j can no longer be queried at e_j and would need re-deriving, e.g. via a recipient- or population-level mean-TTO query, with unclear risk of double-counting against ϕ_j.

- G **re-calibration.** Any TTO-blended Σ shifts the scale of $s_j(t)$ relative to the original X-based Σ; the confidence modifier G likely needs re-tuning under any TTO variant, and may explain part of the Binary-to-TTO underperformance observed in Section 4.3.
- **Asymmetric deep encoder/decoder.** The two-layer network of Section 3.7 uses matched encoder/decoder depth and width; decoupling them (e.g. a wider encoder with a narrower, more constrained decoder, or vice versa) may better balance representational capacity against the denoising-based regularization, and warrants exploration alongside a wider sweep over masking rate, dropout, and bottleneck width d to address the variance observed in Section 4.3.
- **Time-informed p_j.** The influence score p_j (Section 3.4) currently reflects only whether opening is informative; extending it to reward recipients whose *speed* of opening is informative about others' opening speed — not merely their opening probability — is a natural companion to the TTO constructions of Section 3.6 but has not been formulated or tested here.

References

Žid, Č., Alves, R., and Kordík, P. 2025. Active Recommendation for Email Outreach Dynamics. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*, November 10–14, 2025, Seoul, Republic of Korea. ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3746252.3760832